

Mingjia Huo

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Research Summary

PhD candidate working on **interactive world models and video generation** (action-conditioned, causal, few-step generation for long-horizon rollout) and **VLM video understanding and benchmarking**. Using games as a testbed, my research moved from evaluating how LLM/VLM agents perceive and act in interactive environments (ICLR'26, NeurIPS'26) to building world models that simulate them (ICML Workshop'26, MBZUAI-IFM PAN World Model).

Education

University of California, San Diego (UCSD)

PhD in Electrical and Computer Engineering (GPA: 4.0)

- Advisor: Hao Zhang, Tajana Rosing

California, US

Sep 2023 - Jun 2027

University of Illinois, Urbana-Champaign (UIUC)

MS in Electrical and Computer Engineering (GPA: 3.95)

- Advisor: Kirill Levchenko

Illinois, US

Aug 2020 - Dec 2022

Peking University

BS in Computer Science (Turing Class, GPA: 3.76, rank top 10%)

- Advisor: Qun Huang

Beijing, China

Sep 2016 - Jun 2020

Industry Research Experience

MBZUAI - Institute of Foundation Models

AI Research Intern | *Real-time causal interactive world model on PAN*

California, US

Jun 2026 - Present

- Led the design and implementation of **causal video generation** for the PAN world model, using diffusion forcing and teacher forcing for chunk-by-chunk interactive rollout.
- Distilled multi-step causal models into **4-step generators** with self-forcing, enabling minute-scale rollout with Infinite-RoPE and accumulated KV-cache history.
- Collaborated with the infra team to deploy **Video Sparse Attention** on H200 machines, retaining only 15% of attention blocks for remote history while keeping dense attention on recent history.
- Trained action-conditioned video generation models on gameplay videos with diverse action and prompt sets.

Google DeepMind - Gen AI

Student Researcher | *VLM spatiotemporal reasoning in Atari* → **[1] NeurIPS'26, first author**

California, US

Jun 2025 - Oct 2025

- Developed an **Atari-based benchmark** to evaluate high-FPS video understanding and decision-making of VLMs.
- Designed an agentic workflow that dynamically adjusts the gameplay context history.
- Identified key VLM failure modes in visual perception, motion detection, text-history overreliance and long-context reasoning.

Research Experience

Action-Conditioned World Model for Third-Person Games

Hao AI Lab, UCSD → **[2] ICML Workshop'26, first author**

Jan 2026 - Jun 2026

- Built an action-conditioning module on Wan2.1-1.3B and fine-tuned it on 250K Zelda gameplay videos, achieving consistent action following while preserving visual fidelity.
- Distilled a 40-step bidirectional teacher into a 4-step causal student with **diffusion forcing and self-forcing**, and proposed adaptive forcing, which progressively extends student rollout under teacher guidance.
- Improved robustness on out-of-distribution scenes by scaling action embeddings and attention-map injection.
- Contributed the action-conditioned world model to the open-source FastVideo  framework (~**4.5K GitHub stars**).

LMGame-Bench and Multi-Turn RL for LLM Agents in Games

Hao AI Lab, UCSD → **[3] ICLR'26, co-first author**

Feb 2025 - Oct 2025

- Benchmarked state-of-the-art LLMs on six video games with and without structured harnesses for game-state understanding, and established the LMGame-Bench  leaderboard (~**1K GitHub stars**).
- Built LLM agents with visual perception and memory modules for interacting with game environments.
- Led multi-turn RL training of Qwen2.5-7B-Instruct on Sokoban and Tetris with our open-source GRL  system, improving generalization on spatial reasoning, planning and multi-turn decision-making.

Learnable LLM Watermarking

UCSD → [4] **ICML'24, first author**; [5] **Cell Reports (under revision), first author**

Sep 2023 - Feb 2025

- Trained adaptive controllers to set token-level watermark strength from token history, with a multi-objective optimization framework balancing detectability and semantic coherence.
- Improved the Pareto frontier of the detection-semantics trade-off on the C4 dataset.
- Proposed MedMark, which extends learnable watermarking to medical LLMs via a split-and-preference network, outperforming prior methods on medical QA and radiology report summarization in both LLM-judge and expert clinical evaluation.

Publications

[1] Mingjia Huo, Yao Fu, Bo Chang, et al. Evaluating Spatiotemporal Reasoning of Vision-Language Models in Atari Gameplay. **NeurIPS'26**.

[2] Mingjia Huo, Kaiqin Kong, Minshen Zhang, et al. Robust Action-Conditioned World Modeling for Third-Person Games. **ICML Workshop'26**.

[3] Lanxiang Hu*, Mingjia Huo*, Yuxuan Zhang, et al. LMGame-Bench: How Good are LLMs at Playing Games? **ICLR'26**.

[4] Mingjia Huo, Sai Ashish Somayajula, Youwei Liang, et al. Token-Specific Watermarking with Enhanced Detectability and Semantic Coherence for Large Language Models. **ICML'24**.

[5] Mingjia Huo*, Ruiyi Zhang*, Avi Patel, et al. Learnable Watermarking Enables Traceable Medical Text Generation with Large Language Models. **Cell Reports**, under revision.

[6] Mingjia Huo, Sean Perez, Linda Awdishu, et al. AI-Driven Tacrolimus Dosing in Transplant Care: Cohort Study. **JMIR AI'25**.

[7] Mingjia Huo, Maxwell Bland, Kirill Levchenko. All Eyes on Me: Inside Third-Party Trackers' Exfiltration of PHI from Healthcare Providers' Online Systems. **WPES'22**.

Skills

Generative video diffusion, causal generation, distillation, action conditioning

LLM / VLM multi-turn RL, VLM benchmarks and evaluation

Infra video sparse attention, sequence parallel, KV-cache rollout

Programming Python, PyTorch, C/C++

Selected Awards

2019 **Fellowship**, Hui-Chun Chin and Tsung-Dao Lee Chinese Undergrad Research Endowment

2015 **Silver Medal**, Chinese Mathematical Olympiad

2015 **Gold Medal**, Chinese Girls' Mathematical Olympiad